

Research Question

When a user trades an ERC-20 stablecoin for ETH on a DEX protocol, they must first grant delegate authority to the DEX's Permit2 contract account over their tokens via ERC-20 `approve()`. This approval persists indefinitely unless explicitly revoked. The study asks: what is the measurable economic cost of this approval mechanism, how do users actually behave around it, and does that behavior vary across user activity levels and across time?

Background

Every user who interacts with a DEX must delegate spending authority to a smart contract, often for an unlimited amount. This study examines what it costs to approve (gas fees paid to issue approvals) and how users manage it. The ETH → stablecoin pair is used because it represents a clear, economically motivated action. Additionally, by examining approval behavior across both DEX swapping (Uniswap, SushiSwap) and lending (Aave, Compound), the study can also test whether the trust delegation dynamics differ meaningfully across protocol types. Users who supply or borrow may have stronger incentives to manage their approvals carefully — or they may be even more passive given the complexity involved.

Study Scope & Parameters

Chains: Ethereum + Polygon

Time Period: ****INCLUDE NOVEMBER!!!** (November 2025, December 2025, January 2026, and February 2026)

- December/January boundary could provide some insights into tax-loss harvesting behavior (users realizing losses at year-end tend to swap more aggressively in late December, which may spike approvals)
- February represents a fairly “normal” month

DEX Swapping:

- Uniswap V3 (Should be V4)
- SushiSwap V2

Lending:

- Aave V3 (supply and borrow)
- Compound V3 (supply and borrow)

Token Pairs and Actions

Uniswap V3 and SushiSwap — swap: ETH → stablecoin (directional; user selling ETH to acquire a stable asset)

On Ethereum:

- ETH → USDC
- ETH → USDT
- ETH → DAI

On Polygon:

- POL → USDC
- POL → USDT
- POL → DAI

*Are we interested in using WETH as well? (Likely No for now)

Aave V3:

Supply: ETH deposited as collateral into the Aave V3 pool on Ethereum. These are the dominant collateral assets and each requires its own approve () to the Aave pool contract before deposit.

Borrow: USDC/DAI/USDT borrowed against supplied collateral.

*Same process for Polygon, low volume concerns

Compound V3:

Supply: ETH deposited as collateral.

Borrow: USDC/DAI/USDT.

*Same process for Polygon, low volume concerns

Process

Identify qualifying wallets. 1 question here is whether we should include bots?

Ex: <https://etherscan.io/address/0x5b43453fce04b92e190f391a83136bfbecedfd1>

Decile construction. Sort wallets by action count (swap count, supply count, or borrow count) within each month. Assign deciles D1 through D10. Sample from each decile skewing toward upper deciles. Flag wallets at >3 standard deviations above the mean action count as likely bots or automated actors for separate treatment.

Fetch approval history. For each sampled wallet, query `erc20_ethereum.evt_Approval` (via Dune API) to retrieve all `approve()` events emitted during the study window. Capture: block time, token address, spender address, allowance amount, gas used, gas price.

Classify approvals. For each approval event, classify as: unlimited (allowance = `uint256 max`), exact-amount (allowance matches a specific transaction value), or revocation (allowance = 0).

Compute metrics per wallet, aggregate by decile and month.

Key Metrics

Action count — swaps, supplies, or borrows per wallet per month. Establishes the activity distribution and decile structure.

Approval count and type — total `approve()` calls per wallet in the window, broken down by unlimited vs. exact-amount vs. revoke.

Approval-to-action ratio — approvals divided by qualifying actions per wallet. Near 0 means the wallet coasts on a prior unlimited approval. Near 1 means it approves before each action. Above 1 indicates revoke-and-reapprove cycling.

Approval gas cost per wallet — total USD spent on `approve()` transactions during the window

Outstanding allowance at end of period — for wallets that approved but never revoked, the remaining allowance at end of February. For unlimited approvals this is always `uint256 max` regardless of how many actions were executed; this is the persistent exposure users may carry unknowingly.

Expected Findings by Protocol

Uniswap V3: Most wallets will show one approval (or zero if approved before December) followed by multiple swaps with no further approvals. The approval-to-swap ratio for D1–D5 wallets is expected to be close to zero in January and February. High-frequency D9–D10 wallets likely set unlimited approvals well before the study window or use Permit2 sub-approvals, so their in-window approval count may appear near zero despite hundreds of swaps.

SushiSwap: Because SushiSwap requires a fresh exact-amount approval before each swap, the approval-to-swap ratio should track much closer to 1:1 across all deciles. This provides a contrast with Uniswap — same economic action, very different approval mechanics and therefore very different accumulated trust exposure. SushiSwap's active user population is also much smaller.

Aave: Supply and borrow represent the most complex approval flows in the study. Supplying collateral requires approving the collateral token to the Aave pool. Borrowing does not require an approval on receipt, but repayment does. Users maintaining active positions across the study window may show approval behavior tied to position management rather than new interactions.

Compound: Initial findings from DUNE api queries show Compound's active borrower population is small (345 wallets in January) relative to the DEX arms, meaning the lending arms will have smaller but potentially more analytically interesting samples where individual wallet behavior is more visible.

Additional Points that Could Be Explored

Mevbots

Flashbots rose because Proof-of-Stake changed validator economics. PoS validators are known in advance (the validator set is public), so MEV-Boost and block builder/proposer separation became the dominant architecture. This created a new layer of infrastructure — relays, block builders, searchers — that users now implicitly trust every time they submit a transaction.

Maybe by identifying transactions in the dataset with `max_priority_fee_per_gas = 0` that appear in the public mempool anyway, those could be examples of this leak. It would be a small finding, but could be related to users who attempted to protect themselves from MEV but failed. However this may not be enough evidence, probably need to explore a stronger way of proving this.